Reg. No. | | | | | | | | | | [12EE81]

EIGHTH SEMESTER B.E EXAMINATION, JUNE - 2016

POWER SYSTEM OPERATION AND CONTROL

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	a.	Write briefly about SCADA. With the help of digital computer configuration, explain the controlling and monitoring of power system.	07
	b.	Explain with a neat block diagram the dual computer configuration of the supervisory control and Data Acquisition (SCADA) system.	07
2.	a.	With neat schematic and block diagram, explain the DC excitation system for Automatic voltage Regulator (AVR).	07
	b.	Obtain the mathematical model of speed governing system for steam turbine.	07
3.	a.	Derive the expression for tie-line power and frequency deviation for two area system.	07
	b.	Two areas are interconnected, the generating capacity of area A is 36000MW and regulating characteristics are 1.5% of capacity per 0.1Hz. Area D has a generating capacity of 4000 MW and its regulating characteristic are 1% of capacity per 0.1Hz. Find the frequency deviation for the areas share of 400MW disturbance accruing in area D and the resulting tie-line flow.	05
	c.	Explain the concept of tie line bias control.	02
4.	a.	What is meant by sub-synchronous resonance?	02
	b.	Explain different sources of reactive power generation and absorption in Electrical power system.	05
	c.	Derive the equation to get the relation between voltage, power and reactive power at a node.	07
5.	a.	Explain the problem of unit commitment. What are the constraints in solving the unit commitment problem?	06
	b.	Write the algorithm for priority list method of solving unit commitment problem.	08
6.	a.	Define power system security and explain major functions involved in the system security.	07
	b.	With the help of flow chart, explain the contingency analysis.	07
7.	a.	Explain the role of linear sensitivity factors in the contingency analysis.	07
	b.	Explain the contingency selection procedure with help of flow chart.	07

8. a. Derive the expression for availability and unavailability in terms of failure and repair rate for two state generator model. **07**

b. A power plant has generating units with the data shown in the table below determine the capacity outage probability table and indicate the cumulative probabilities. **07**

Unit No	Capacity (MW)	Failure rate/Year (λ)	Repair rate/year (μ)
1	100	1.0	19.0
2	150	0.1	4.9
3	250	1.1	73.0